

PGNCOM GAS BILLING DISCOVERY PROTOTYPE BLUEPRINT

Pain Point Driven | Business Process | Requirements | Architecture | ERD | Use Cases

Positioning

Prototype ini adalah discovery instrument untuk memperlihatkan secara seamless bagaimana pain point billing gas ditangani dari master data, decision and calculation, validation, sampai invoice eDoc. Prototype tidak memisahkan mana capability yang sudah tersedia atau harus dibangun; seluruh kebutuhan ditampilkan sebagai satu target experience. Hasil discovery menjadi requirement dan build backlog untuk implementasi berikutnya di MyDX.

Document Scope	Included
Business scope	Gas billing calculation-to-invoice untuk PGNCOM sebagai billing executor.
Prototype modules	Master Data & Scenario Configuration; Billing Decision & Calculation; Generate Invoice eDoc.
Primary objective	Membuktikan treatment setiap pain point, mengonfirmasi business rule, data, control, dan output yang dibutuhkan.
Downstream boundary	Posting/distribution/collection dapat direpresentasikan sebagai output interface, tetapi bukan fokus prototype ini.

Updated 31 July 2026

DOCUMENT STRUCTURE

Section	Content
1. Background, Pain Point & Proposed Solution	Context, problem statement, proposed prototype and pain point-to-solution alignment.
2. Business Process	End-to-end gas billing process from regulatory setup and gas usage to invoice and adjustment.
3. Requirement Detail & Structured Research	Detailed functional, data, rule, control and output requirements; research is embedded under the requirement it supports.
4. Logical Architecture	Seamless prototype architecture from source data to business outputs.
5. Conceptual ERD	Main master, configuration and transaction entities and their relationships.
6. Use Cases & Prototype Resolution Flow	Pain-point-driven scenarios showing input, steps, decisions, outputs and acceptance evidence.

Reading principle

Setiap requirement dan use case selalu ditarik kembali ke pain point pada Section 1. Requirement yang tidak mendukung pain point, business process, atau output prototype sengaja dihilangkan.

1. BACKGROUND, PAIN POINT & PROPOSED SOLUTION

1.1 Background

PGNCOM berperan sebagai billing operation executor. PGN/business owner tetap menjadi pemilik customer, contract/PJBG, kebijakan harga, eligibility, regulatory interpretation, commercial approval, dan accounting policy. Karena itu prototype berfokus pada controlled execution: menarik data, menentukan treatment berdasarkan master dan rule yang disetujui, menghitung, memvalidasi, menyimpan trace, serta menghasilkan invoice eDoc.

Karakter billing gas tidak hanya ditentukan oleh volume dikali harga. Nilai tagihan dipengaruhi oleh periode efektif regulasi dan harga, customer/contract eligibility, supply source, allocation/quota, unit conversion, formula, payment scheme, billing frequency, recurring/non-recurring treatment, adjustment, dan approval. Kompleksitas utama adalah memastikan keputusan yang benar digunakan secara konsisten dan dapat dijelaskan.

Core problem statement

Billing team membutuhkan cara yang terstruktur, traceable, dan configurable untuk mengubah regulatory/commercial policy menjadi billing decision, calculation, approval, dan invoice - tanpa bergantung pada spreadsheet, re-entry, atau pengetahuan satu orang.

1.2 Captured Pain Points

ID	Pain Point	Business Problem	Prototype Treatment
P01	Repeated recalculation	Revisi tarif, prorata, estimasi, atau koreksi memicu hitung ulang berulang dan hasil dapat tidak konsisten.	Old-vs-new simulation, calculation version, delta and adjustment trace.
P02	Tariff period uncertainty	Sulit menentukan cut-off dan periode harga ketika perubahan berlaku di tengah periode atau disetujui terlambat.	Effective-dated price, split-period logic, retroactive flag and period trace.
P03	Multiple price & formula	Harga dan formula berbeda menurut customer, contract, segment, source, volume, atau commercial arrangement.	Formula catalogue, parameter master, decision priority and contract assignment.
P04	Regulatory change	Regulasi/keputusan harga dapat diamendemen, dicabut, diganti, atau terlambat diterima.	Regulation hierarchy, status, scope, version activation and impact simulation.
P05	Fragmented source data	Data customer, contract, meter, tariff, historical bill dan invoice tersebar sehingga rawan re-entry dan mismatch.	Scenario/data loader, mapping, completeness check, data snapshot and lineage.
P06	Different payment schemes	Postpaid, prepaid dan hybrid membutuhkan treatment balance, deposit, AR dan residual invoice yang berbeda.	Payment scheme configuration, balance application priority and residual billing.
P07	UOM, allocation & component complexity	Currency, UOM, conversion, allocation/quota, eligible/excess volume dan price component meningkatkan risiko calculation error.	Conversion profile, allocation split, component calculation and rounding trace.
P08	Manual validation & approval	Pengecekan dan approval manual tidak konsisten, sulit dipantau, dan dapat menjadi bottleneck.	Validation rule, exception worklist, maker-checker, SLA and approval evidence.
P09	Weak explainability & audit trail	Sulit menjelaskan kenapa angka tagihan terbentuk dan versi data/rule apa yang dipakai.	Why This Result?, immutable snapshot, intermediate values and version history.
P10	Billing period & charge treatment inconsistency	Recurring/non-recurring charge serta harian, mingguan, bulanan,	Billing frequency/calendar, engagement type, trigger, proration

ID	Pain Point	Business Problem	Prototype Treatment
		semesteran, tahunan atau on-demand tidak konsisten antar kontrak.	and grouping.
P11	Invoice inconsistency	Invoice tidak selalu transparan terhadap calculation breakdown, supporting data dan adjustment history.	Controlled invoice template, charge breakdown, source reference and document version.

1.3 Proposed Solution

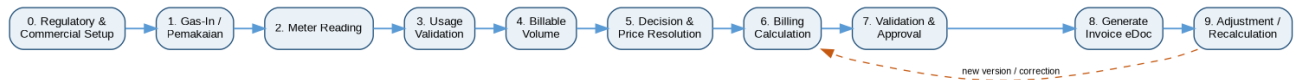
Module	Purpose	Key User Experience	Primary Output
A. Master Data & Scenario Configuration	Menyiapkan seluruh commercial, regulatory, pricing, formula, frequency, payment dan invoice configuration secara effective-dated dan versioned.	User memilih atau memuat scenario, melihat source reference, status, effective date dan dependency.	Approved configuration and repeatable scenario data.
B. Billing Decision & Calculation	Menentukan eligibility, applicable price, formula, parameter, payment treatment, usage split dan calculation result.	User menjalankan calculate/simulate, melihat intermediate values, comparison, exception dan Why This Result?.	Calculation header/line, decision trace, variance, exception and approved result.
C. Generate Invoice eDoc	Menyusun approved calculation menjadi commercial invoice yang konsisten dan transparan.	User preview, validate breakdown, approve, generate PDF and create adjustment when needed.	Invoice eDoc, document metadata, invoice lines and adjustment reference.

Prototype success criterion

Untuk setiap pain point, user harus dapat memilih scenario, memuat data, memahami decision, menjalankan calculation/simulation, melihat exception dan approval, menghasilkan output, lalu menyatakan rule tersebut Confirmed, Need Clarification, atau Gap. Tampilan menu saja tidak dianggap menyelesaikan pain point.

2. BUSINESS PROCESS

2.1 End-to-End Gas Billing Process



Proses dimulai sebelum gas digunakan, yaitu ketika regulasi, contract/PJBG, price version, formula, allocation, payment scheme dan billing calendar disiapkan. Setelah penggunaan tervalidasi, system menyelesaikan decision dan price resolution sebelum menghitung. Recalculation tidak menerima hasil lama; ia menghasilkan calculation version dan adjustment trace baru.

2.2 Detailed Business Process

Step	Business Activity	Key Input	Primary Output	Pain Points
0	Regulatory & Commercial Setup	Regulation, PJBG/addendum, customer eligibility, payment scheme, frequency, formula, price version, allocation.	Approved and effective configuration.	P02, P03, P04, P06, P10
1	Gas-In / Pemakaian	Customer, service/delivery point, meter, supply source, usage period.	Raw usage event / consumption period.	P05, P07
2	Meter Reading	Opening/closing reading, timestamp, reading type, meter status, source.	Raw metered volume.	P05, P07
3	Usage Validation	Prior reading, tolerance, estimation/correction method, conversion profile.	Validated usage and validation status.	P05, P07, P08
4	Billable Volume	Validated volume, UOM, conversion factor, eligible allocation, excess policy.	Billable quantity per pricing bucket.	P02, P07
5	Decision & Price Resolution	Customer/contract scope, regulation, supply source, location, effective dates, rule priority.	Applicable regulation, price version, component and formula.	P02, P03, P04, P09
6	Billing Calculation	Billable quantity, rate, fixed/variable component, formula parameter, payment scheme, rounding.	Calculation header, lines and intermediate values.	P01, P03, P06, P07, P09, P10
7	Validation & Approval	Prior bill, variance threshold, exception reason, override rule, approval matrix.	Approved, warning, rejected or pending result.	P08, P09
8	Generate Invoice eDoc	Approved calculation, invoice template, numbering, due-date rule, supporting breakdown.	Invoice preview/PDF and document metadata.	P11
9	Adjustment / Recalculation	Original calculation/invoice, new price/formula/usage version, approval reference.	New calculation version, delta and debit/credit adjustment proposal.	P01, P02, P04, P09, P11

2.3 Role Boundary

Role	Primary Responsibility in Process	Prototype Implication
PGN / Business Owner	Owns commercial and regulatory interpretation, contract, price policy, eligibility, allocation, approval and accounting/tax policy.	Master/rule change must carry source reference, owner and approval.

Role	Primary Responsibility in Process	Prototype Implication
PGNCOM / Billing Executor	Loads data, runs calculation, validates exception, prepares invoice, retains evidence and follows approved policy.	Worklist, decision trace, calculation, exception and eDoc must be operationally usable.
Prototype	Executes approved configuration and makes every decision/result visible.	No silent fallback or overwritten history; all assumptions are visible for discovery.

3. REQUIREMENT DETAIL & STRUCTURED RESEARCH

Section ini adalah requirement baseline utama. Setiap requirement menyebut pain point yang didukung, riset/evidence yang menjadi dasar, data yang dibutuhkan, rule yang harus dijalankan, output yang harus muncul di prototype, serta bukti penerimaan yang harus dapat dilihat dalam discovery session.

3.1 Research Basis Used in Requirements

ID	Research / Source Basis	Requirement-Relevant Insight
S1	PGNCOM Invoice-to-Cash proposal deck	Customer/contract/usage/tariff/historical billing sources; billing intake, calculation, validation and document generation.
S2	Client/workshop slide and clarification	Gas-in, meter recording, repeated recalculation, tariff-period uncertainty, multiple pricing/formulas, postpaid/prepaid/hybrid and sudden price changes.
R1	Perpres 40/2016 as amended by Perpres 121/2020	Gas price determination and HGBT framework; parent-amendment relationship.
R2	Permen ESDM 15/2022	Procedure for HGBT users/prices in industry; revokes Permen ESDM 8/2020; user, supplier, contract volume, plant gate and evaluation data.
R3	Permen ESDM 15/2021	Organization/work procedure regulation; not a gas-price rule and must not be used as pricing basis.
R4	Kepmen ESDM 91/2023 amended through Kepmen 281/2026	Industry HGBT user/price list and amendment history.
R5	Kepmen ESDM 77/2025 amended through Kepmen 284/2026	HGBT for public electricity supply.
R6	BPH Migas pricing framework and location-specific decisions	Customer class, location/network, seller, tariff and review cycle for household/small customer context.
R7	Gas digital integration context	Daily volume validation, monitoring, reporting and billing integration reinforce need for dated usage and source trace.

Regulatory research note

Riset yang dikompilasi sampai 31 July 2026 mengoreksi tiga hal penting dari slide workshop: Perpres 121/2020 adalah perubahan atas Perpres 40/2016; Permen ESDM 8/2020 telah dicabut oleh Permen ESDM 15/2022; dan Permen ESDM 15/2021 bukan regulasi harga gas. Prototype wajib menyimpan status, parent/amendment, replacement and source reference agar tarif obsolete tidak terpakai.

RQ-01 Regulatory & Commercial Configuration

Linked pain points: P02, P03, P04, P09

Aspect	Requirement Construction
Business objective	Mengubah regulasi, keputusan harga dan commercial interpretation yang telah disetujui menjadi configuration yang efektif, versioned dan dapat ditelusuri.
Research / evidence basis	S2, R1-R6. Regulasi dapat memiliki parent/amendment, revoked/replacement status, customer/industry/location/supply scope, price list, allocation and effective date. Informasi regulasi tidak boleh disimpan hanya sebagai PDF tanpa structured fields.
Functional requirement	Maintain regulation header, hierarchy, status, effective period, authority, source document, scope, affected customer/segment/location/supply source, interpretation summary, approval reference and linked price policy/version. Provide impact view when a new version is activated.
Master data / input	Regulation Master; Regulation Scope; Authority; Source Document; Parent/Amendment/Replacement; Effective From/To; Interpretation Owner; Approval Reference; Affected Policy/Customer/Contract.
Business rules & controls	Inactive/revoked regulation cannot be selected; overlapping active scope is blocked; new version does not overwrite old version; interpretation is business-owned; calculation uses only approved active mapping.
Prototype output	Regulation detail, current consolidated view, impacted contracts/periods, selected regulatory basis in Decision Trace, and change-impact simulation.
Acceptance evidence	User can activate an approved amendment, see prior version preserved, identify impacted records, and explain why the new or old regulation is selected for a sample date.
Discovery confirmation needed	Actual authority, approval workflow, mapping granularity, effective-date convention, retroactive treatment and list of applicable gas-price regimes.

RQ-02 Customer, Contract, Service & Meter Context

Linked pain points: P03, P05, P06, P07, P10

Aspect	Requirement Construction
Business objective	Memastikan calculation selalu memiliki customer, contract/PJBG, service/delivery point, meter and commercial context yang tepat.
Research / evidence basis	S1-S2. Deck identifies customer and contract systems plus usage/metering; gas billing can vary by segment, location, source, delivery point and contract. HGBT decisions may identify eligible users, supply and contract volume.
Functional requirement	Maintain customer and segment; contract period/addendum; service/delivery point; supply source; meter assignment; currency; payment scheme; billing frequency; billing engagement; formula/price assignment history. Validate active period and mandatory dependency before run.
Master data / input	Customer Master; Contract/PJBG; Contract Version/Addendum; Service/Delivery Point; Supply Source; Meter Master; Tax Profile; Currency; Contact/Invoice Address.
Business rules & controls	One contract may contain multiple services and billing engagements; a meter cannot be ambiguously active for multiple services in the same period; expired/inactive contract blocks calculation; addendum creates a new effective version.
Prototype output	Customer-contract-service profile, active configuration timeline, missing dependency flags and source lineage.
Acceptance evidence	For a selected billing period the system resolves exactly one active contract/service/meter context and blocks calculation when a mandatory relationship is missing or overlapping.
Discovery confirmation needed	Actual contract hierarchy, one-to-many relationships, supply source mapping, meter replacement handling and source-system ownership.

RQ-03 Usage Intake, Meter Validation & Billable Volume

Linked pain points: P02, P05, P07, P08

Aspect	Requirement Construction
Business objective	Menghasilkan billable volume yang valid, traceable and ready for price resolution.
Research / evidence basis	S2 and R7. Workshop shows gas-in followed by meter recording and initial validation. Readings may be actual, estimated or corrected. Different units and allocations require controlled conversion and bucket split.
Functional requirement	Upload/sync meter reading; map meter to service; show opening/closing reading and timestamps; classify actual/estimated/corrected; detect missing, duplicate, negative, rollover and abnormal variance; apply approved correction/estimation; convert raw usage to billing UOM; split eligible and excess volume.
Master data / input	Usage Header/Line; Meter Reading; Reading Type; Meter Status; Prior Reading; Validation Rule; Estimation/Correction Method; Conversion Profile; Heating/Energy Factor where applicable; Allocation/Quota.
Business rules & controls	Invalid usage cannot silently continue; every correction retains original reading and reason; conversion factor is effective-dated and source-referenced; split method uses actual granularity when available or approved prorata rule.
Prototype output	Validated usage, correction history, billable volume per bucket, validation status and exception reason.
Acceptance evidence	Prototype can run actual, estimated, corrected and abnormal samples; user can trace raw reading to billable volume and see any blocked/warning condition.
Discovery confirmation needed	Actual raw/billing units, conversion factor source, tolerance, estimation duration, rollover rule, daily/monthly granularity and approval owner.

RQ-04 Tariff / Price, Version, Component, UOM & Allocation

Linked pain points: P01, P02, P03, P04, P07, P09

Aspect	Requirement Construction
Business objective	Menentukan harga yang benar berdasarkan regime, scope, period, component and allocation without hard-coded values.
Research / evidence basis	S2, R1-R6. Gas prices may be regulated or commercial, customer/location/source-specific, effective-dated and componentized. Research identified Rp/m3, USD/MMBTU and BBTUD allocation contexts; tax inclusion may vary by component.
Functional requirement	Maintain price policy and version; effective from/to; currency and UOM; customer/segment/location/supply scope; component type; unit rate; tax inclusion flag; allocation/quota; excess policy; approval; previous/replacement version. Detect gaps and overlap.
Master data / input	Price Policy; Price Version; Price Component; Currency/UOM; Conversion Profile; Allocation; Eligible/Excess Policy; Tax Inclusion; Source/Approval.
Business rules & controls	Applicable price is resolved by date and scope; overlapping equal-priority versions block run; missing price creates exception; allocation split generates separate calculation lines; component rounding follows approved profile.
Prototype output	Applicable price lines, eligible/excess split, component-level charge basis, price source/effective date and alternative rejection reason.
Acceptance evidence	User can demonstrate HGBT eligible/excess, mid-period price change, missing price and overlapping price; result shows exactly which version/component is applied.
Discovery confirmation needed	Full price catalogue, component definitions, UOM/currency conversion, tax/FX/rounding policy, allocation period and commercial fallback treatment.

RQ-05 Formula Master & Formula Parameter

Linked pain points: P01, P03, P06, P07, P10

Aspect	Requirement Construction
Business objective	Mendukung formula yang dapat dipilih dan diubah melalui configuration, bukan hard-coded per customer.
Research / evidence basis	S2 confirms multiple price types and formulas. Formula must separate expression from parameter values/sources so the same formula can work across contracts and versions.
Functional requirement	Maintain formula ID/name/type/expression/version/status; define required parameters, source, data type, default/override permission and rounding; test formula with sample inputs; assign formula through Billing Rule; preserve previous formula version.
Master data / input	Formula Master; Formula Version; Formula Parameter; Parameter Source; Test Case; Validation Result; Owner/Approval.
Business rules & controls	Only approved active formula can execute; missing required parameter blocks run; parameter override requires reason/authorization; formula version used by historical calculation is immutable.
Prototype output	Selected formula, parameter values and source, intermediate values, calculated components, test result and formula version in Decision Trace.
Acceptance evidence	User can select different formula through contract/rule, execute fixed, usage, fixed+usage, minimum/take-or-pay, allocation/excess, prorata and adjustment examples without changing source code.
Discovery confirmation needed	Actual formula catalogue, allowed operators/functions, formula ownership, testing/approval method, parameter hierarchy and exception formulas.

RQ-06 Billing Rule, Eligibility & Decision Resolution

Linked pain points: P02, P03, P04, P06, P09, P10

Aspect	Requirement Construction
Business objective	Menentukan treatment yang berlaku dengan urutan keputusan yang konsisten dan explainable.
Research / evidence basis	The main billing complexity is deciding which regulation, price, formula and treatment applies. Contract, segment, source, location, period, volume, payment scheme and engagement type can influence the decision.
Functional requirement	Maintain condition, scope, priority/precedence, effective date, linked price version, formula, payment/frequency treatment and exception path. Resolve candidates, select one, explain why alternatives are rejected and show fallback only when explicitly configured.
Master data / input	Billing Rule; Rule Condition; Rule Priority; Engagement Assignment; Formula ID; Price Version ID; Regulation/Contract Scope; Effective Period; Fallback; Exception Action.
Business rules & controls	Highest valid priority wins; conflicting same-priority rules block calculation; no silent default; rule evaluation snapshot is stored; manual override is controlled and visible.
Prototype output	Decision trace: evaluated conditions, eligible candidates, selected rule, rejected alternatives, final price/formula/payment treatment and reason.
Acceptance evidence	For two contrasting contracts, user can demonstrate different selected formulas/prices and explain the exact condition and priority that caused the decision.
Discovery confirmation needed	Actual precedence among regulation, contract/addendum and commercial circular; fallback policy; manual override authority; scope granularity and exception route.

RQ-07 Payment Scheme: Postpaid, Prepaid & Hybrid

Linked pain points: P06, P09, P11

Aspect	Requirement Construction
Business objective	Menerapkan treatment settlement yang sesuai untuk postpaid, prepaid and hybrid without confusing it with billing frequency.
Research / evidence basis	Client explicitly identified postpaid, prepaid and hybrid. These are best modelled as Payment/Settlement Scheme because they determine when payment is applied relative to usage and whether a residual invoice or balance movement is created.
Functional requirement	Maintain payment scheme; deposit/top-up source; opening balance; application priority; credit/debit treatment; residual invoice rule; refund/shortfall treatment; ledger movement; due-date applicability.
Master data / input	Payment Scheme Master; Customer/Contract Assignment; Balance Ledger; Deposit/Top-up Transaction; Application Priority; Credit Limit; Residual Invoice Rule.
Business rules & controls	Postpaid creates receivable/invoice after usage; prepaid applies available balance and blocks/warns on shortfall according to policy; hybrid applies prepaid portion first then creates residual invoice or other approved treatment.
Prototype output	Charge, balance application, remaining balance, residual amount, invoice/deposit treatment and movement trace.
Acceptance evidence	Prototype can compare the same usage under postpaid, prepaid and hybrid and reconcile charge, balance movement and final invoice amount.
Discovery confirmation needed	Actual balance ledger source, deposit ownership, top-up timing, application priority, insufficient balance, refund, credit limit and accounting treatment.

RQ-08 Billing Frequency, Calendar & Recurring / Non-Recurring Engagement

Linked pain points: P02, P10, P11

Aspect	Requirement Construction
Business objective	Membangun periode billing, trigger charge and invoice grouping consistently per contract.
Research / evidence basis	Daily, weekly, monthly, quarterly, semester, annual and on-demand are Billing Frequency values. Langganan/non-langganan are better represented as Billing Engagement or Charge Recurrence: recurring, non-recurring/one-time, event-based and true-up.
Functional requirement	Maintain frequency, anchor date, meter cut-off, calculation date, approval cut-off, invoice date, due-date rule; engagement type; recurring flag; charge trigger; active period; proration; grouping/separation; annual component schedule and event-based adjustment.
Master data / input	Billing Frequency Master; Billing Calendar; Billing Engagement; Charge Type; Trigger; Proration Rule; Grouping Rule; Due-Date Rule.
Business rules & controls	Frequency and payment scheme are separate dimensions; recurring charge generates according to calendar; non-recurring requires event/reference; inactive period is not billed; mid-cycle start/end uses approved proration; grouping follows contract policy.
Prototype output	Constructed billing period, scheduled worklist, charge instances, proration detail and invoice grouping.
Acceptance evidence	Prototype can generate monthly recurring usage+fixed fee, a one-time adjustment and an annual component with correct period and invoice line grouping.
Discovery confirmation needed	Actual frequencies, calendars, cut-offs, due dates, proration convention, holiday treatment, charge trigger and grouping across services/contracts.

RQ-09 Billing Calculation, Simulation & Recalculation

Linked pain points: P01, P02, P03, P04, P06, P07, P09, P10

Aspect	Requirement Construction
Business objective	Menghasilkan calculation yang repeatable, versioned and transparent, serta memungkinkan simulasi tanpa mengubah historical result.
Research / evidence basis	Workshop pain is repeated recalculation and sudden price change. Calculation must support normal run, split-period, old-vs-new simulation, retroactive recalculation, affected-record identification and adjustment delta.
Functional requirement	Create billing run and calculation snapshot; resolve rule/price/formula; execute line calculations; show intermediate values; compare prior vs new; identify impacted period/customer; create new version; calculate delta; propose adjustment linked to original invoice.
Master data / input	Billing Run; Calculation Header/Line; Input Snapshot; Decision Trace; Prior Calculation; Calculation Version; Variance; Adjustment Proposal.
Business rules & controls	Simulation never changes approved history; recalculation creates new version; mass impact requires controlled selection and approval; totals reconcile to lines; rounding is deterministic; adjustment references original result/invoice.
Prototype output	Approved calculation, old-vs-new comparison, variance reason, affected record list, calculation version, delta and adjustment proposal.
Acceptance evidence	User can reproduce repeated recalculation and mid-period/retroactive price scenarios; old result remains visible and new result explains the delta.
Discovery confirmation needed	Actual mass-run volume, recalculation triggers, performance/SLA, debit-credit policy, grouping and approval threshold.

RQ-10 Validation, Exception & Approval

Linked pain points: P05, P07, P08, P09, P11

Aspect	Requirement Construction
Business objective	Memisahkan straight-through result dan exception secara konsisten serta memastikan approved result memiliki evidence.
Research / evidence basis	Pain points include manual checking, inconsistent approval and weak audit trail. Validation domains span customer/contract, meter/usage, regulation/price, formula/parameter, variance, balance and invoice reconciliation.
Functional requirement	Configure block/warning rules; create exception worklist; assign owner; capture correction/override reason and evidence; route maker-checker or multi-level approval; track SLA/aging; freeze approved snapshot; prevent invoice generation for blocked result.
Master data / input	Validation Rule; Exception Type/Severity; Threshold; Owner; Evidence; Override Permission; Approval Matrix; Decision; Timestamp; SLA.
Business rules & controls	Block vs warning is explicit; override is role-restricted; rejected result cannot generate invoice; approval refers to a specific calculation version; any change after approval creates a new version.
Prototype output	Validation summary, exception list, owner/action, approval route, decision evidence, aging and approved version.
Acceptance evidence	Prototype can block missing price, show abnormal usage warning, route an exception, capture approval and prove invoice is generated only from approved calculation.
Discovery confirmation needed	Thresholds, role matrix, approval levels, SLA, override policy, evidence retention and escalation.

RQ-11 Generate Invoice eDoc & Adjustment Document

Linked pain points: P01, P09, P11

Aspect	Requirement Construction
Business objective	Mengubah approved calculation menjadi invoice yang konsisten, transparent and linked to its source calculation.
Research / evidence basis	Proposal deck identifies document generation. Client pain requires clear breakdown and adjustment trace. Commercial invoice eDoc should be designed separately from tax/e-Faktur integration.
Functional requirement	Assemble invoice header and line from approved result; controlled template; invoice numbering/version; issue/due date; customer/contract/service; usage and price breakdown; currency/tax fields; supporting attachment; document status; adjustment/debit-credit reference; PDF preview/generation.
Master data / input	Invoice Header; Invoice Line; Template; Numbering Rule; Due-Date Rule; Calculation Reference; Supporting Document; Document Version/Status; Adjustment Link.
Business rules & controls	Invoice total reconciles to approved calculation; one calculation version cannot silently produce duplicate issued invoice; template version is stored; cancellation/adjustment preserves history; tax document reference is separate configuration.
Prototype output	Invoice preview/PDF, document metadata, charge breakdown, source/decision reference, status history and adjustment document.
Acceptance evidence	User can trace each invoice line to calculation line and source rule/price; total reconciles; regenerated preview uses same approved snapshot; adjustment links to original invoice.
Discovery confirmation needed	Actual template, mandatory fields, invoice number, due date, signature, attachments, tax/PPN/e-Faktur linkage, language and distribution payload.

RQ-12 Pain-Point-Driven Prototype UX & Discovery Evidence

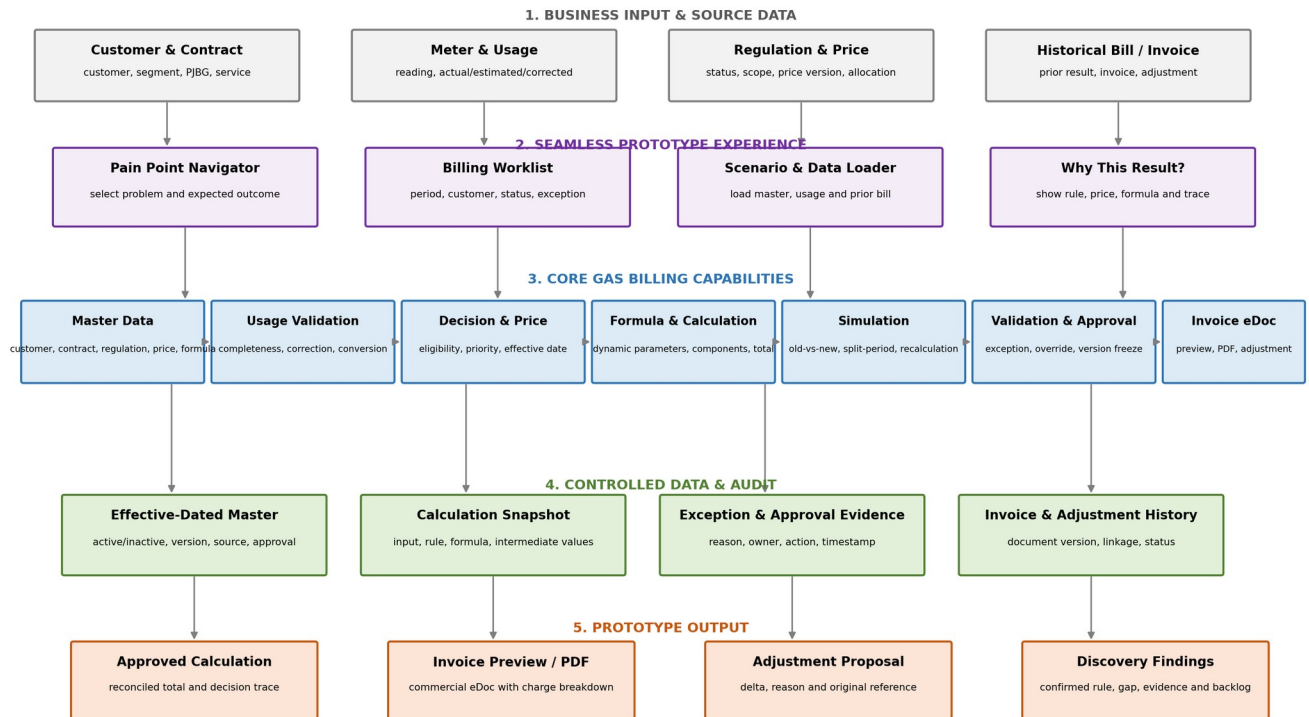
Linked pain points: All P01-P11

Aspect	Requirement Construction
Business objective	Membuat prototype mudah digunakan untuk membuktikan penyelesaian pain point dan menangkap requirement build-ready.
Research / evidence basis	A menu-driven demo risks losing the problem context. The prototype should start from a recognized pain point and guide the user through evidence, decision, calculation, exception, approval and output.
Functional requirement	Pain Point Navigator; guided scenario; scenario/data loader; Billing Worklist; Decision Trace/Why This Result?; calculation comparison; exception/approval; invoice preview; discovery finding capture; repeatable synthetic/masked datasets.
Master data / input	Pain Point; Scenario; Expected Outcome; Evidence; Requirement IDs; Test Data; Expected Result; Discovery Status; Open Question; Owner; Backlog Note.
Business rules & controls	Each scenario links to a pain point and requirement; user must see input-decision-result-control-output; findings are marked Confirmed/Need Clarification/Gap; synthetic/masked data is used unless approved otherwise.
Prototype output	Repeatable scenario acceptance pack, confirmed rules, open questions, evidence gaps and prioritized build backlog for MyDX implementation.
Acceptance evidence	In one guided session, user can select a pain point, reproduce it, understand the result, validate expected outcome and capture a concrete follow-up requirement.
Discovery confirmation needed	Priority pain points, personas, maximum acceptable steps, evidence owners, acceptance authority and discovery governance.

4. PROTOTYPE LOGICAL ARCHITECTURE

Arsitektur menyajikan satu seamless prototype. Pengguna tidak melihat pemisahan antara feature yang sudah tersedia dan custom component; mereka melihat satu alur problem-to-output. Pada tahap implementasi berikutnya, hasil discovery digunakan untuk menentukan build detail di MyDX.

PROTOTYPE LOGICAL ARCHITECTURE



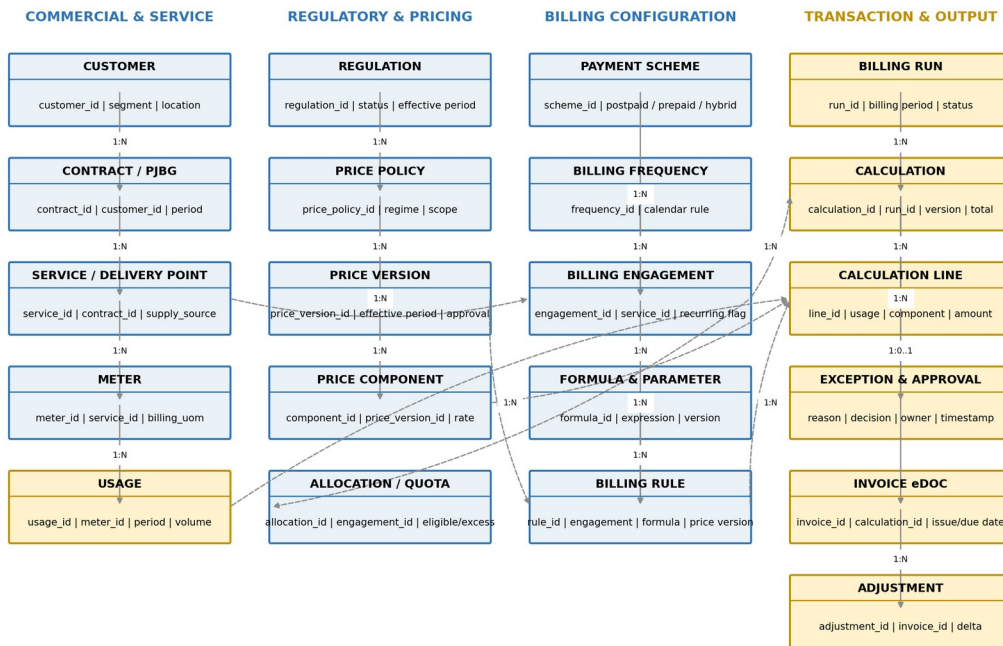
Cross-cutting controls: role-based access, maker-checker, effective date, versioning, audit trail, source reference, and immutable calculation history.

Layer	Responsibility	Key Principle
Business Input	Customer/contract, meter/usage, regulation/price and historical billing/invoice.	Every record carries source, period and owner.
Seamless Experience	Pain Point Navigator, worklist, scenario loader and Why This Result?.	User starts from a business problem, not a technical menu.
Core Capabilities	Master data, usage validation, decision/price resolution, formula/calculation, simulation, approval and eDoc.	All decisions are configuration-driven and explainable.
Controlled Data & Audit	Effective-dated master, snapshots, exception/approval evidence and document history.	History is immutable; changes create new versions.
Prototype Output	Approved calculation, invoice PDF, adjustment proposal and discovery findings.	Output is both business evidence and future build requirement.

5. CONCEPTUAL ERD

ERD berikut berfungsi untuk menyepakati data object dan relationship utama. Physical database design, datatype, index and normalization ditentukan setelah rule, volume and integration source tervalidasi.

CONCEPTUAL ERD - GAS BILLING PROTOTYPE



Billing Rule is the central assignment object: it connects a billing engagement to the selected formula and effective price version. Transaction history is never overwritten; recalculation creates a new calculation version and, when approved, an adjustment linked to the original invoice.

Relationship	Cardinality / Meaning
Customer - Contract - Service - Meter - Usage	One customer may have multiple contracts; contract may have multiple services; service may have multiple meter histories; meter produces usage records.
Regulation - Price Policy - Price Version - Price Component	Regulatory/commercial basis produces effective-dated price versions and component rates.
Payment Scheme / Billing Frequency - Billing Engagement	An engagement defines how and when a service is billed: settlement scheme, recurrence, trigger, calendar and grouping.
Formula & Parameter - Billing Rule - Price Version	Billing Rule is the assignment/decision object connecting engagement, formula and applicable price version with condition and priority.
Billing Run - Calculation - Calculation Line	A run contains calculations; lines retain usage, component, quantity, rate, amount and trace.
Calculation - Exception / Approval - Invoice - Adjustment	Only approved calculation generates invoice; later changes create a new calculation version and adjustment linked to the original invoice.

6. USE CASES & PROTOTYPE RESOLUTION FLOW

Setiap use case dimulai dari pain point pada Section 1 dan menggunakan requirement/output pada Section 3. Flow standar prototype adalah: Select Pain Point -> Load Scenario & Evidence -> Resolve Decision -> Calculate/Simulate -> Validate & Approve -> Generate Output -> Capture Discovery Finding.

Use-case acceptance rule

Skenario dinyatakan berhasil apabila prototype memperlihatkan input, decision/rule, calculation/workflow, exception/control, approved output and traceability. Hasil yang hanya berupa angka tanpa alasan dan source tidak dianggap selesai.

UC-01 Repeated Recalculation after Tariff Revision

Use Case Element	Detail
Pain point	P01
Requirements used	RQ-01, RQ-04, RQ-06, RQ-09, RQ-10, RQ-11
Scenario setup / required data	Original approved calculation/invoice; new price version; effective date; approval reference; usage history.
Prototype resolution flow	1) Select repeated recalculation. 2) Load original bill. 3) Activate approved new price version in simulation. 4) Resolve impacted period and records. 5) Run old-vs-new calculation. 6) Review delta and reason. 7) Approve new calculation version. 8) Generate adjustment proposal linked to original invoice.
Expected prototype outputs	Original result remains unchanged; new calculation version; affected-record list; old-vs-new variance; adjustment amount and trace.
Discovery acceptance	Billing owner confirms delta, affected period, split method and approval treatment.

UC-02 Mid-Period Price Change & Cut-Off

Use Case Element	Detail
Pain point	P02
Requirements used	RQ-01, RQ-03, RQ-04, RQ-06, RQ-08, RQ-09
Scenario setup / required data	Old/new price versions; effective date; meter/billing cut-off; daily usage or approved proration rule.
Prototype resolution flow	1) Select tariff period uncertainty. 2) Load usage and two price versions. 3) Show cut-off timeline. 4) Split usage by actual daily records or approved prorata. 5) Calculate separate price-period lines. 6) Show retroactive flag if approval is late. 7) Validate and preview invoice.
Expected prototype outputs	Transparent period split, quantity per period, rate per line, total and decision trace.
Discovery acceptance	Business owner confirms which date governs price, the split method and retroactive rule.

UC-03 Formula Decision by Contract / Segment

Use Case Element	Detail
Pain point	P03
Requirements used	RQ-02, RQ-04, RQ-05, RQ-06, RQ-09
Scenario setup / required data	Two customer/contracts, formula catalogue, parameter values, price components and rule priority.
Prototype resolution flow	1) Select multiple formula problem. 2) Load two contracts. 3) Evaluate eligibility and rule priority. 4) Select formula and parameters for each. 5) Calculate. 6) Open Why This Result? to show selected and rejected alternatives.
Expected prototype outputs	Different but explainable results by contract; formula/parameter version; intermediate values and reason.
Discovery acceptance	Billing owner confirms formula catalogue and precedence; any missing condition becomes a requirement.

UC-04 Regulation Amended / Revoked / Missing Mapping

Use Case Element	Detail
Pain point	P04
Requirements used	RQ-01, RQ-04, RQ-06, RQ-09, RQ-10
Scenario setup / required data	Base regulation, amendment/replacement, active/revoked status, affected scope and linked price version.
Prototype resolution flow	1) Select regulatory change. 2) Load regulatory hierarchy. 3) Attempt calculation using revoked version. 4) System blocks and explains. 5) Activate approved replacement mapping. 6) Simulate impact. 7) Re-run and retain source reference.
Expected prototype outputs	Blocked obsolete rule, approved replacement, impacted customers/periods and regulatory decision trace.
Discovery acceptance	Business/legal owner confirms interpretation and mapping; system proves it only executes approved configuration.

UC-05 Fragmented or Incomplete Source Data

Use Case Element	Detail
Pain point	P05
Requirements used	RQ-02, RQ-03, RQ-10, RQ-12
Scenario setup / required data	Customer, contract, meter, usage and price files with missing/duplicate/mismatched records.
Prototype resolution flow	1) Select fragmented data. 2) Load sample sources. 3) Map key fields. 4) Run completeness and relationship checks. 5) Show exceptions by owner. 6) Correct or confirm source. 7) freeze calculation input snapshot.
Expected prototype outputs	Billing worklist with completeness status, mismatches, source lineage, correction and unresolved exceptions.
Discovery acceptance	Data owners confirm source, field meaning, correction authority and integration requirement.

UC-06 Postpaid vs Prepaid vs Hybrid

Use Case Element	Detail
Pain point	P06
Requirements used	RQ-02, RQ-07, RQ-09, RQ-11
Scenario setup / required data	Same usage/charge; payment scheme; opening balance, top-up/deposit, application priority and residual rule.
Prototype resolution flow	1) Select payment scheme. 2) Run postpaid: create full invoice/receivable basis. 3) Run prepaid: apply balance and show remaining balance/shortfall. 4) Run hybrid: apply balance first then calculate residual invoice. 5) Compare results and ledger trace.
Expected prototype outputs	Charge, balance application, residual invoice, remaining balance and movement trace for each scheme.
Discovery acceptance	Commercial/accounting owner confirms deposit, residual, insufficient balance and refund treatment.

UC-07 HGBT Allocation, Excess Volume, UOM & Components

Use Case Element	Detail
Pain point	P07
Requirements used	RQ-03, RQ-04, RQ-05, RQ-06, RQ-09
Scenario setup / required data	Validated usage; billing UOM/conversion; HGBT eligibility; allocation/quota; regulated and excess commercial price components.
Prototype resolution flow	1) Select allocation/UOM problem. 2) Load raw usage and conversion profile. 3) Convert to billing UOM. 4) split eligible and excess. 5) apply component prices. 6) calculate and round. 7) show quantity-rate-amount trace.
Expected prototype outputs	Eligible and excess calculation lines, conversion evidence, component charges, total and rounding trace.
Discovery acceptance	Owner confirms allocation period/source, conversion factor and excess treatment.

UC-08 Manual Validation, Exception & Approval

Use Case Element	Detail
Pain point	P08
Requirements used	RQ-03, RQ-04, RQ-09, RQ-10
Scenario setup / required data	Normal result plus missing price, abnormal usage or high variance samples; validation thresholds and approval roles.
Prototype resolution flow	1) Select manual approval problem. 2) Run validation. 3) Straight-through item passes. 4) Exception item is blocked/warned. 5) Assign owner and evidence. 6) correct/override with reason. 7) route maker-checker approval. 8) freeze approved version.
Expected prototype outputs	Validation summary, exception worklist, SLA/owner, correction/override evidence and approval history.
Discovery acceptance	Process owner confirms severity, thresholds, roles, SLA and evidence requirements.

UC-09 Why This Result? & Audit Trace

Use Case Element	Detail
Pain point	P09
Requirements used	RQ-01 to RQ-11
Scenario setup / required data	Any approved calculation with customer, contract, regulation, rule, price, formula, parameter, usage and approval data.
Prototype resolution flow	1) Open a calculation or invoice. 2) Select Why This Result?. 3) Display source snapshot, eligibility, candidate rules, selected price/formula, parameter values, intermediate calculation, override and approval. 4) compare with previous version when available.
Expected prototype outputs	Complete decision and calculation trace that explains every invoice line without opening code or spreadsheet.
Discovery acceptance	Billing/audit users confirm required trace granularity and retention.

UC-10 Recurring, One-Time and Different Billing Frequencies

Use Case Element	Detail
Pain point	P10
Requirements used	RQ-02, RQ-05, RQ-08, RQ-09, RQ-11
Scenario setup / required data	Monthly recurring usage, recurring fixed fee, one-time adjustment, annual component; billing calendar and grouping rule.
Prototype resolution flow	1) Select recurring/frequency issue. 2) Build billing period from calendar. 3) generate recurring charge instances. 4) trigger one-time event. 5) apply proration if service starts/ends mid-cycle. 6) group or separate invoice lines according to contract.
Expected prototype outputs	Correct billing period, charge instances, proration and invoice line grouping for monthly/annual/one-time components.
Discovery acceptance	Contract/billing owner confirms frequency, trigger, proration, cut-off and grouping.

UC-11 Controlled Invoice eDoc Generation

Use Case Element	Detail
Pain point	P11
Requirements used	RQ-09, RQ-10, RQ-11
Scenario setup / required data	Approved calculation, invoice template, numbering and due-date rule, customer/contract/service and supporting details.
Prototype resolution flow	1) Select invoice inconsistency. 2) open approved calculation. 3) assemble invoice preview. 4) reconcile header total to lines. 5) show usage/price breakdown and source reference. 6) approve/generate PDF. 7) create adjustment document if scenario requires.
Expected prototype outputs	Consistent invoice PDF/eDoc, document version/status, calculation linkage, supporting breakdown and adjustment reference.
Discovery acceptance	Billing/commercial users confirm layout, mandatory fields, total reconciliation, numbering, due date and supporting attachment.

Final prototype narrative

Prototype harus dapat didemokan dari pain point ke business outcome: problem dipilih, evidence dimuat, decision dijelaskan, calculation/simulation dijalankan, exception dan approval terlihat, invoice/adjustment dihasilkan, lalu rule atau gap dicatat. Dengan konstruksi ini, prototype tetap intuitif untuk discovery dan hasilnya langsung dapat diterjemahkan menjadi backlog implementasi MyDX.